**3** Fig. 3.1 shows a portable shower used on a campsite. The bag is filled with water. The water is heated using infrared radiation from the Sun.

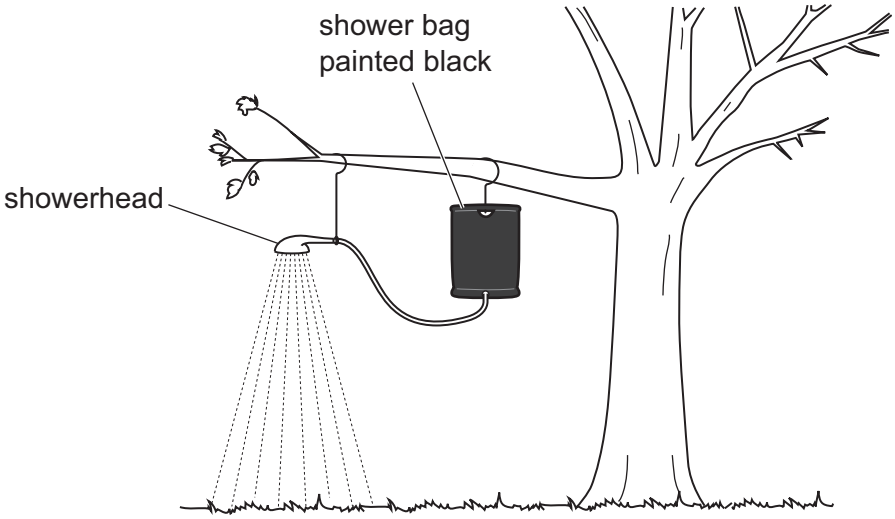


**Fig. 3.1**

**(a) (i)** Explain why the shower bag is painted black.

.....

..... [1]

**(ii)** Explain a **disadvantage** of radiation from the Sun being the only source to heat the water.

.....

..... [1]

**(b)** Solar energy is a renewable energy resource.

State **two other** renewable energy resources.

1 .....

2 .....

[2]

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**(c)** During the day, the Sun shines on the shower bag and some of the energy in the infrared radiation from the Sun transfers to the thermal energy stores of the water.

The water absorbs 60% of the energy incident on the bag. The temperature of the water rises from 10 °C to 43 °C.

The mass of the water in the bag is 40 kg. The specific heat capacity of water is 4200 J / (kg °C).

Calculate the energy incident on the shower bag during the day.

Show your working.

energy = ..... [4]

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